

EE/CprE/SE 491 WEEKLY REPORT 1

Start Date: 2/6/2026 – End Date: 2/17/2026

Group number: sddec26-07

Project title: Socialdeck

Client &/Advisor: Dr. Maruf

Team Members/Role:

Thabang Pila: Student Lead, Backend

Phuong Tran: Team Coordination, Project Management & Frontend

Ibrahim Aldulaimi: Documentation, Design Review, Code Review & Backend

Nathan Ervin: Assignment Coordination, Turn-In & Frontend

Joshua Reynolds: Code Review & Backend

Laura Lopez-De la Cruz: Assignment Coordination, Turn-In & Frontend

- **Weekly Summary:**

This week the primary objective was to establish a clear understanding of the team structure and how frontend and backend development will collaborate throughout the project. As this is Sprint 1, the focus was on defining sub-teams and clarifying the role each member will play moving forward. Initial research began on the technologies and tools that will be used to better understand how individual responsibilities integrate within the overall codebase. The team also discussed establishing consistent breakout meeting times to improve coordination and task delegation. In addition, contribution expectations were outlined, including participating in issue discussions, creating pull requests, and managing merges within the repository. Overall, the week emphasized learning, organization, and preparation for structured development in the upcoming sprint.

- **Past Week Accomplishments:**

The primary goal for the past week was onboarding and becoming comfortable with the existing codebase. The team reviewed Figma Dev Mode to better understand how design components translate into Flutter implementation. The technology stack, including Flutter, Firebase, and Riverpod, was discussed to establish a foundational understanding of the project's architecture. Each team member successfully cloned the repository and ran the project locally to verify proper environment configuration. A team contract was created and finalized, and sub-teams were formed based on individual preferences for frontend and backend development. The existing Flutter

structure and repository organization were explored to gain familiarity with the application's architecture. Overall, the week focused on onboarding, understanding the system structure, and preparing for implementation.

- **Thabang Pila:** Created and organized team communication channels for project coordination. Worked with Phuong to coordinate team availability and confirm the first onboarding session. Led onboarding by walking members through project background, codebase structure, and frontend and backend workflow. Created and posted technical setup documentation, onboarding forms, and resource videos in Discord to help members set up local environments and run the repository. Prepared weekly presentation content to give more detail on the existing codebase and development process. Worked with Phuong Tran to send out role preference forms and helped finalize role assignments. Introduced the team to Sprint 1 workflow and initial contribution expectations.
- **Phuong Tran:** Cloned project codebase in GitHub to personal computer. Downloaded necessary applications and dependencies such as Android Studio, Flutter, Dart, etc. Set up the local environment and test run the project on my computer. Watched instructional videos and documents related to the project and technologies involved. Formulated questions, created and sent out a Google Form to gather member preferences, and analyzed each member's strength, weakness, and preferences to determine role assignment and task delegation. Oversee GitHub Project board and plan for individuals's tasks for upcoming Sprint 1. Prepared weekly presentation to update and inform members of meeting contents, goals, and upcoming tasks.
- **Ibrahim Aldulaimi:** Reviewed and refined the team contract to ensure clarity in role definitions, decision-making policies, collaboration standards, and documentation expectations. Participated in onboarding sessions focused on understanding the overall project architecture, including frontend and backend layer separation and Clean Architecture principles used within the repository. Explored the existing Flutter structure to better understand how UI components connect to state management using Riverpod and how data flows between presentation, domain, and data layers. Reviewed Figma Dev Mode to understand design requirements and how visual components translate into Flutter widgets. Contributed to discussions regarding Scrum workflow organization within the GitHub Projects section, including issue tracking, sprint structure, and contribution expectations. Spent additional time reviewing repository structure, dependency management, and understanding how state management integrates with Firebase services within the current implementation.
- **Nathan Ervin:** Set up working environment for project, including setting up local Visual Studio and updating the necessary applications for this project, including Dart, Flutter, etc. Cloned the project GitHub into this environment and began simple testing and investigating the project structure and formatting. Spent time investigating the front-end development screens on Figma, and did research into how to translate Figma screens and pages into actual front-end code that can be used in the project. Investigated GitHub commit reports to see how the project has developed and how it plans to evolve. Spent time with teammates, went through project onboarding and development issues to ensure that all team

members are on the same page for the beginning of this project. Refreshed my skills in several of the languages being used in this project, so that we can hit the ground running instead of needing several weeks to get back up to speed with the language.

- **Joshua Reynolds:** Cloned, ran, and looked over the repository on own machine to familiarize with a small part of the codebase before onboarding. Participated in the onboarding meeting. Started research with provided materials into Riverbase as well as the preferred method of architecture for the project. Spent time looking into how Firebase handles security and how that will impact the connection between the backend code and the Firebase server. Looked into separation of code within the current project repository to partly understand how the separation of concerns of the code impacts design choices made in the project.
- **Laura Lopez-De la Cruz:** Successfully cloned and ran the repository on my personal device for both iOS and Android to ensure I had a proper start to the project. Participated in onboarding sessions with our team to get a better understanding of the project architecture and the tech stack we will be using. Looked into the Github Project dashboard tutorial to learn how the flow works in a team setting. Began personal research into Flutter and the IDE Cursor I will be using and looking into the many features it has. Assisted in reviewing our team contract to ensure all areas were completed and submitted it for our group prior to our due date.
- **Pending issues:**
 - **Thabang Pila:** A key pending issue has been transitioning from solo development to onboarding five new developers. Most of the existing codebase was written months earlier, so codebase review and documentation are still ongoing to make handoffs clearer for the team. The main challenge has been translating prior implementation decisions into onboarding material that is easier for new contributors to follow. To address this, earlier learning resources were revisited and broken into simpler step by step guidance so members can enter the codebase gradually without feeling overwhelmed. This remains an active effort while the team moves deeper into Sprint 1.
 - **Phuong Tran:** For frontend onboarding, there will be a steep learning curve due to the knowledge each frontend developer needs to acquire. The developer needs to become familiar with the Figma setup and the templates built already from our UI/UX designer. Good knowledge and experience with Flutter is necessary in addition to Widgets and OOP. Each developer also needs to learn the existing codebase and database before making progress on top of the existing work. Each frontend developer needs to learn to make new template(s) that is applicable and reusable for a variety of screens. The biggest challenge is learning how to use new tools and how to implement them into the current workflow.
 - **Ibrahim Aldulaimi:** During onboarding, the backend architecture and implementation of Clean Architecture principles presented a significant learning

curve. The layered structure, including presentation, domain, and data separation, required deeper analysis to understand system responsibilities and data flow. Particular attention has been directed toward understanding how Riverpod state management integrates with Firebase services and how state transitions propagate across layers. The overall depth of the existing implementation initially felt overwhelming; however, continued review of the codebase, instructional videos, and external tutorials has gradually improved comprehension. Learning and architectural familiarization remain ongoing as preparation for meaningful backend contributions in the upcoming sprint.

- **Nathan Ervin:** For frontend development, there is a new exposure to Figma, which I have not used prior to this project, that I need to spend more time gaining experience and familiarity with. There is also a need to become very familiar with the existing codebase and to understand the vision and format on which the original code is based. Overall, the biggest challenge I see going forward is acclimating to new technologies, relearning these languages, which I have not used for quite some time, and familiarizing myself with the existing code structure.
- **Joshua Reynolds:** Working on internalizing the principles of the architecture, but requires more hands-on experience to fully understand the practice of Clean Architecture. Requires more time to dive into the codebase at a more intimate level to understand some of the separation of responsibilities present in the code. Needs to focus on the difference between the data, domain, providers and service layers. Had difficulties with interpreting Figma designs. Difficulties with interpretation and understanding of Frontend design decisions and application can hinder the design decisions made by the backend team, slowing down the cycle of development.
- **Laura Lopez-De la Cruz:** Prior to cloning the repository to my local device, I had minor issues with dependencies I was missing. I was able to resolve them by using the Flutter installation page and resources from our resources page. As a frontend developer for this project there is a vast amount of technology that needs to be understood prior to starting. Ensuring that we start our research on our tech stack and reaching out to others in our team will become crucial as we continue to advance. We also need to ensure familiarity with the current codebase and structure to avoid confusion when developing as a team.

- **Individual contributions:**

<u>NAME</u>	<u>Individual Contributions</u> (Quick list of contributions. This should be short.)	<u>Hours this week</u>	<u>HOURS cumulative</u>
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Thabang Pila	● Organized Discord coordination channels ● Guided codebase and workflow walkthrough ● Posted setup resources for local environment configuration ● Helped finalize role assignments ● Introduced Sprint 1 workflow	14	14
Phuong Tran	● Formulate questionnaires ● Send out Google Form ● Gather team members's preferences and delegate roles ● Compose weekly meeting powerpoint slides ● Watch instructional videos and read documentations ● Set up and test run local environment for project	7	7
Ibrahim Aldulaimi	● Refined and organized team contract ● Participated in onboarding and backend architecture review ● Analyzed layered structure and state management flow ● Reviewed documentation and instructional materials to strengthen backend understanding	6	6
Nathan Ervin	● Watched instructional videos to strengthen understanding of architecture ● Spent time working on understanding the code within our GitHub ● Set up and ran a local environment with the project. ● Investigated Figma UI/UX Development Ideas	6	6
Joshua Reynolds	● Watched instructional videos,	4	4

	• Dove into the existing codebase to prepare for future contributions. • Set up and ran the current project on a local machine to test the environment.		
Laura Lopez-De la Cruz	• Watched instructional videos to ensure understanding of the projects architecture • Set up and did a test run on my local environment to ensure that application was able to run on both iOS and Android • Spent time trying to understand technologies we will be using on the project.	6	6

- **Comments and Extended Discussion:**

The team maintained strong communication during onboarding and role assignment discussions. Clear expectations for responsibilities and collaboration standards were established early in the project lifecycle. Scheduling alignment required coordination due to differing academic workloads; however, consistent communication through Discord and GitHub supported smooth onboarding. Overall team dynamics have been constructive and focused on long-term project organization.

- **Team Structure for Upcoming Weeks:**

For the remainder of the sprint cycle, work will be organized into three implementation duos. Each duo will include one frontend developer and one backend developer working collaboratively on the same selected screen from Figma to ensure consistent integration between UI, state management, and backend services. This structure is intended to promote end-to-end feature ownership while maintaining alignment between design, application logic, and data handling.

Each pair will review the selected design screen, define expected behaviors, user interactions, edge cases, and error states, and then divide responsibilities accordingly. Frontend developers will focus on translating Figma designs into Flutter widgets, structuring layouts, and implementing Riverpod state handling for loading, success, empty, and error states. Backend developers will focus on identifying required data structures, defining service interactions, analyzing state flow, and implementing the backend logic or placeholders necessary to support the screen's functionality.

This duo-based model will serve as the foundation for coordinated development in

upcoming weeks.

This structure represents the current development plan and may be adjusted in future sprints as project requirements, workload distribution, and team familiarity with the architecture evolve.

- **Focus for the Upcoming Week:**

The primary objective for the upcoming week is to solidify sub-team structure and clarify individual responsibilities. Each team member will ensure a clear understanding of their assigned role within frontend or backend development and how that role contributes to the overall architecture.

Additional time will be dedicated to researching relevant technologies, tools, and architectural patterns to strengthen familiarity with the existing codebase. Sub-teams will establish consistent breakout meeting times to coordinate task delegation and define contribution expectations.

Frontend developers will begin implementation by selecting relatively simple screens from Figma to establish a development workflow and improve familiarity with the project structure. Emphasis will be placed on incremental learning, state handling patterns, and understanding integration points rather than full feature completion at this stage.

- **Summary of Weekly Advisor Meeting:**

No advisor meeting was held during this reporting period, however communication with the project advisor has been ongoing. The team has been conducting weekly meetings with the clients to discuss project direction, expectations, and development planning. An advisor meeting will be scheduled to review overall progress and sprint alignment.